

Convention: power rails above components; GND below.

The schematic shows the display side of the Volthium reader. Key components include:

- Power Regulation:** A 2.2uF/25V capacitor (C1) and a 10uF capacitor (C2) are used for power filtering. A 78E3.3-0.5 voltage regulator (U4) provides a 3.3V supply.
- Microcontroller:** The ESP32-S3-WROOM-1-N16R8 (U1) is the central component, featuring PSRAM and various pins for communication and power.
- Display Interface:** The EPD_FFC_24 display (U2) is connected via SPI (SPK_MOSI, SPK_MISO, SPK_CS) and I2C (I2C_SDA, I2C_SCL).
- Connectors:** A USB-C connector (J1) provides power and data. A USB-A connector (J3) is used for debugging (DBG_UART_TX, DBG_UART_RX, USB_OTG). A USB-C connector (J4) is used for power (PWR_FLAG, GND).
- Status Indicators:** Several status LEDs (V12_CAT5E, V12_PROT, V12_PROT, V12_PROT, V12_PROT) are connected to the microcontroller.